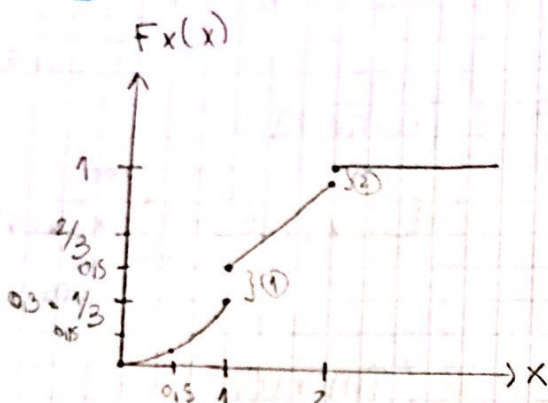


GUÍA (3)

(17)

31) Sea X v.a con $F_X(x) = \frac{x^3}{3} \mathbb{I}_{\{0 \leq x < 1\}} + \frac{2x+1}{6} \mathbb{I}_{\{1 \leq x < 2\}} + \mathbb{I}_{\{x \geq 2\}}$.



MIXTA

a) $E[X]$

b) $E[X | X < 1]$ y $E[X | X \leq 1]$.

a) $E[X] = \sum_{x \in A} x p(x=x) + \int_I x F_X'(x) dx$

$I: I_1(0, 1) + I_2(1, 2)$

$A: 1 \rightarrow 1/6 \quad 2 \rightarrow 1/6$

$$E[X] = 1 \cdot \frac{1}{6} + 2 \cdot \frac{1}{6} + \int_0^1 x \cdot x^2 dx + \int_1^2 \frac{2}{6} x dx =$$

$$E[X] = \frac{1}{6} + \frac{2}{6} + \left. \frac{x^4}{4} \right|_0^1 + \left. \frac{2x^2}{2} \right|_1^2 = \frac{5}{4}$$

b) $E[X | X < 1] = \frac{E[X \mathbb{I}_{\{X < 1\}}]}{P(X < 1)}$

$$= \frac{\int_0^1 x^3 dx}{F(1) - P(1)} = \frac{1/4}{1/2 - 1/6} = \frac{3}{4}$$

NOTA

ESPERANZA $\rightarrow$ valor esperado

Discreta: $A = \{x: p(x=x) > 0\}$ $\nearrow$ átomos

$E[X] = \sum_{x \in A} x \cdot p(x)$

$E[g(x)] = \sum_{x \in A} g(x) p(x)$

$g(x)$ es v.a.

Continua:

$E[X] = \int_{-\infty}^{+\infty} x f_X(x) dx$

$E[g(x)] = \int_{-\infty}^{+\infty} g(x) f_X(x) dx$

Mixta

$E[X] = \sum_{x \in A} x p(x) + \int_I x F_X'(x) dx$
 $\downarrow$
 I
 $\neq$ intervalos

PROP

$E[X]$ es lineal:

$E[X+Y] = E[X] + E[Y]$

$a E[X] = E[aX]$

$X \geq 0 \rightarrow E[X] \geq 0 \quad E[a] = a$

$E[X+a] = E[X] + a$

$$E[X | X \leq 1] = \frac{E[X \cdot 1_{\{X \leq 1\}}]}{P(X \leq 1)}$$

$$= \frac{\int_0^1 x^3 dx + 1 \cdot \frac{1}{6}}{\frac{1}{2}} = \frac{\frac{1}{4} + \frac{1}{6}}{\frac{1}{2}} = \frac{5}{6}$$

3.2 Sea X una v.a. con F. del 2.2.

$$a) E[X] = (-2) \cdot \frac{1}{3} + 2 \cdot \frac{1}{3} + \int_{-2}^1 0 dx + \int_{-1}^1 \frac{1}{6} x dx = \frac{1}{6} \left. \frac{x^2}{2} \right|_{-1}^1 = 0$$

$$b) E[X | |X| = 2] = \frac{E[X \cdot 1_{\{X=2 \cup X=-2\}}]}{P(X=2 \cup X=-2)} = \frac{-2 \cdot \frac{1}{3} + 2 \cdot \frac{1}{3}}{\frac{2}{3}} = 0$$

3.3 Pascal $\rightarrow E[X] = \frac{k}{p}$

3.4 Sea $(\Omega, \mathcal{A}, P) \rightarrow \Omega = \{1, 2, 3, 4, 5, 6\}$

$$\mathcal{A} = \{\emptyset, \{1, 2\}, \{3, 4\}, \{5, 6\}, \{1, 2, 3, 4\}, \{1, 2, 5, 6\},$$

$$\{3, 4, 5, 6\}, \Omega\} \text{ y } P(\{1, 2\}) = \frac{1}{6}$$

$$P(\{5, 6\}) = \frac{1}{2}$$

$$P(\{3, 4\}) = \frac{1}{3}$$

a) Definir X v.a. / $X: \Omega \rightarrow \mathbb{R} / \{-1, \frac{1}{2}\} \subset X(\Omega) \quad \forall E[X] = 0$.

V.A. numérica

discreta

$$E[X | X \in B] = \frac{E[X \cdot 1_{\{X \in B\}}]}{P(X \in B)}$$

continua

$$E[X | X \in B] = \frac{\int_{-\infty}^{+\infty} x f_X(x) 1_{\{X \in B\}} dx}{P(X \in B)}$$

mixta

Planteas esperanzas y prob y socos según info

Ω	$X(\Omega)$
1	-1
2	-1
3	$-x_0$
4	0
5	0
6	$-1/2$

$$E[X] = -1 \cdot \frac{1}{6} + \frac{1}{2} \cdot \frac{1}{3} + x_0 \cdot \frac{1}{2} = 0$$

$$x_0 = 0$$

$$X = -1 \mathbb{1}_{\{\omega \in \{1,2\}\}} + \frac{1}{2} \mathbb{1}_{\{\omega \in \{3,4\}\}}$$

$$b) E[X | X > -1] = \frac{E[X \mathbb{1}_{\{X > -1\}}]}{P(X > -1)} = \frac{\frac{1/6}{1} = 1/5}{\frac{F(-1) - P(X = -1)}{1} = 1/6}$$

$$c) \{-1, 1\} \subset X(\Omega)$$

Ω	$X(\Omega)$
1	-1
2	x_0
3	x_0
4	1
5	1
6	1

$$E[X] = -1 \cdot \frac{1}{6} + x_0 \cdot \frac{1}{2} + 1 \cdot \frac{1}{3} = 0$$

$$x_0 = -1/3$$

$$X = -1 \mathbb{1}_{\{\omega \in \{1,2\}\}} + 1 \mathbb{1}_{\{\omega \in \{3,4\}\}} + \frac{1}{3} \mathbb{1}_{\{\omega \in \{5,6\}\}}$$

$$E[X | X > -1] = 1/5$$

3.5

Discreta

Cant moscas $\rightarrow$ Poisson(10) $\rightarrow \frac{10^x e^{-10}}{x!} = P_X(x)$

$$E[X | X \leq 4] =$$

$$= \frac{E[X \cap \{X \leq 4\}]}{P(X \leq 4)} = \frac{\sum_{i=0}^4 x \cdot P_X}{\sum_{i=0}^4 P_X(x)}$$

Por ser discreta
Sumo q prob puntual

$$= \frac{0 \cdot P(0) + 1 \cdot P(1) + 2 \cdot P(2) + 3 \cdot P(3) + 4 \cdot P(4)}{P(1) + P(2) + P(3) + P(4)}$$

Fórmula de Prob Total

$$E[X] = \sum_i P(X \in A_i) \cdot E[X | X \in A_i]$$

Parto a x en intervalos
o conjuntos disjuntos

$$= e^{-10} \left(\frac{10^1}{1!} + \frac{10^2 \cdot 2}{2!} + \frac{10^3 \cdot 3}{3!} + \frac{10^4 \cdot 4}{4!} \right) =$$

$$= \frac{6830}{3} \div \frac{1930}{3} = \boxed{3,539}$$

CONTINUA

3.6 exponencial media 3, $E[T | T \leq 2]$

$$f_X = \frac{1}{3} e^{-1/3 x}$$

$$3 = 1/\lambda \rightarrow \lambda = 1/3$$

$$E[T] = 3 \dots$$

2º FORMA

Podemos hacer lo
con la
Toulo.
 $\downarrow$
Paramo
integrar

$$E[T | T \leq 2] = \frac{E[T \cap \{T \leq 2\}]}{P(T \leq 2)} = \frac{\int_{-\infty}^{+\infty} x \cdot f(x) dx}{\int_0^2 f_X dx}$$

$$F_X = 1 - e^{-1/3 x} \cap \{x \geq 0\}$$

$$= \frac{\int_0^2 x \cdot \frac{1}{3} e^{-1/3 x} dx}{\int_0^2 \frac{1}{3} e^{-1/3 x} dx} = \frac{0,143}{0,48} = 0,29$$

continua

$$(3.7) a) f_X(x) = \frac{9!}{3!5!} x^3 (1-x)^5 \quad \text{for } \{0 < x < 1\}$$

$$E[X] = \int_0^1 f(x) \cdot x \, dx = \frac{9!}{3!5!} \int_0^1 x^3 (1-x)^5 \, dx$$

$$E[X] = 504 \int_0^1 x^4 - 5x^5 + 10x^6 - 10x^7 + 5x^8 - x^9 \, dx$$

$$E[X] = 504 \left(\frac{1}{5} - \frac{5}{6} + \frac{10}{7} - \frac{10}{8} + \frac{5}{9} - \frac{1}{10} \right) = \boxed{\frac{2}{5}}$$

b) $f_X(x) = \frac{1}{6} x^3 \cdot e^{-x} \quad \text{for } \{x > 0\}$

integrando...

$$E[X] = \int_0^{\infty} \left(\frac{1}{6} x^3 \cdot e^{-x} \right) \cdot x \, dx = \frac{1}{6} \cdot 24 = 4$$

otra forma $\rightarrow \Gamma(4, 1) \rightarrow E[X] = \frac{V}{\lambda} = 4 \checkmark$

↓
valor
esperado gamma de

c) $E[X | X > 1/2] = \frac{E[X \cdot \mathbb{1}_{\{X > 1/2\}}]}{P(X > 1/2)}$

$P(X > 1/2)$

↓
 $1 - P(X \leq 1/2)$

$$E[X | X > 1/2] = \frac{\int_{1/2}^{+\infty} f_X \cdot x \, dx}{1 - \int_0^{1/2} f_X \, dx}$$

$= \frac{4}{0.199} = 4$

↓
 $F(1/2) = \int_0^{1/2} f_X(x) \, dx$

gamma $\rightarrow \frac{\lambda}{(v-1)!} x^{v-1} e^{-\lambda x}$

$v-1=4 \quad \boxed{v=4}$
 $\boxed{\lambda=1}$

38) Sea z una normal estándar, φ su función densidad, $z_0 > 0$

a) Hallar la media $z | z > z_0$ (en función de z_0).

función densidad de z

$$\varphi(z) = \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}z^2}$$

$$E[z | z > z_0] = \frac{E[z \mathbb{1}\{z > z_0\}]}{P(z > z_0)}$$

$$E[z | z > z_0] = \frac{\int_{z_0}^{+\infty} z \cdot \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}z^2} dz}{1 - \Phi(z_0)} = \frac{\frac{\varphi(z_0)}{1 - \Phi(z_0)}}{1 - \Phi(z_0)}$$

b) Deducir que $(1 - \Phi(z_0)) \leq \frac{\varphi(z_0)}{z_0}$

$$\frac{\varphi(z_0)}{1 - \Phi(z_0)} \geq z_0$$

$$1 - \Phi(z_0) \leq \frac{\varphi(z_0)}{z_0}$$

c) $P(z > 4) \leq \frac{\frac{1}{\sqrt{2\pi}} e^{-\frac{4^2}{2}}}{4} = 0,000033$

$$P(z > 4) = \frac{\varphi(4)}{4}$$

$$P(z > 4) = 1 - \Phi(4) = 1 - 0,999967 = 0,000033$$

d) $X = \sigma z + \mu$ $x_0 > \mu$

$$X | X > x_0 = ?$$

$$P(z > 4) = 1 - \Phi(4)$$

$$E[X | X > x_0] = E[\sigma z + \mu | \sigma z + \mu > x_0] =$$

$$\sigma E[z | z > \frac{x_0 - \mu}{\sigma}] + \mu$$

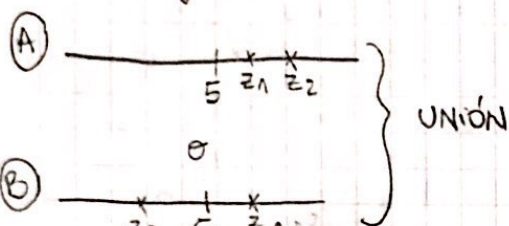
$$\frac{\sigma \varphi(\frac{x_0 - \mu}{\sigma})}{1 - \Phi(\frac{x_0 - \mu}{\sigma})} + \mu =$$

3.9 Sean z_1 y z_2 dos variables normales estandarizadas (no necesariamente independientes)

Demstrar $P(\max(z_1, z_2) > 5) = \frac{2e^{-25/2}}{5\sqrt{2\pi}}$

siempre o alguno > 5
puede ocurrir

$$P(z_1 > 5 \cup z_2 > 5) \leq P(z_1 > 5) + P(z_2 > 5)$$



que alguno sea mayor q' 5

$$\leq \frac{\phi(5)}{5} + \frac{\phi(5)}{5} \leq$$

uso teoremas del 3.8

$$\leq \frac{\frac{1}{\sqrt{2\pi}} e^{-\frac{5^2}{2}}}{5} + \frac{\frac{1}{\sqrt{2\pi}} e^{-\frac{5^2}{2}}}{5}$$

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

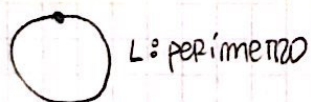
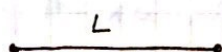
$$P(A \cup B) \leq P(A) + P(B)$$

$$\leq \frac{2 \cdot e^{-25/2}}{5\sqrt{2\pi}}$$

3.10 Se construye un círculo uniendo los extremos de una lambr.

a) $L \sim \exp(\lambda)$ $\frac{1}{\lambda} = 60 \text{ cm} \rightarrow \lambda = 1/60$

Calcular la media del área del círculo.



$$\downarrow \pi r^2$$

$$2\pi r \rightarrow \text{perimetro del círculo.}$$

$$2\pi r = L$$

$$r = L/2\pi$$

$$\text{Área} = \pi \left(\frac{L}{2\pi} \right)^2$$

$$\boxed{A = \frac{L^2}{4\pi}}$$

$x = \text{Área} = g(L) \rightarrow$ el área me quedo como una función de la variable q' yo miro.

continua

$$E[x] = E[g(L)] = \int_{-\infty}^{+\infty} g(L) f_L(L) dL = \Rightarrow \text{sigue}$$

siendo $f_L(L) = \lambda \cdot e^{-\lambda L}$

como el área es $> 0 \rightarrow$ integro de 0 a $\infty \rightarrow$ expo
 edic que el soporte
 $\downarrow$
 de 0 a ∞

$$E[g(L)] = \int_0^{\infty} \frac{L^2}{4\pi} \cdot \frac{1}{60} e^{-L/60} dL =$$

como es difícil de integrar busca
 otra forma

1º como la gamma y la expo
 tienen el **MISMO SOPORTE** $\rightarrow$ lo llevo a una función
 gamma $\Gamma(v, \lambda)$

$$E[g(L)] = \frac{1}{60 \cdot 4\pi} \int_0^{\infty} L^2 e^{-L/60} dL \quad \left\{ \begin{array}{l} f_L(L) = \frac{\lambda^v}{(v-1)!} L^{v-1} e^{-\lambda L} \end{array} \right.$$

$$E[g(L)] = \frac{1}{60 \cdot 4\pi} \int_0^{\infty} L^2 e^{-L/60} \cdot \frac{\left(\frac{1}{60}\right)^3}{2!} dL \cdot \frac{2!}{\left(\frac{1}{60}\right)^3} \quad \left\{ \begin{array}{l} \lambda = 1/60 \\ v-1 = 2 \\ v = 3 \end{array} \right.$$

agrego en termino
multiplicando
y dividiendo

$$E[g(L)] = \frac{1}{60 \cdot 4\pi} \cdot 2! \cdot 60^{3-2} \quad \textcircled{1} = 572,96 \text{ cm}^2 = E[A']$$

LA DENSIDAD INTEGRADA
 en el soporte = 1

$$\int_{\text{sop}} f_X(x) dx = 1$$

2º se que
 $\text{VAR}[L] = E[L^2] - (E[L])^2$

$$E[L^2] = \text{VAR}[L] + (E[L])^2$$

$\downarrow$
 $\frac{1}{\lambda^2}$ $\frac{1}{\lambda}$

$$E[L^2] = 2 \cdot 60^2 \Rightarrow E[g(L)] = E\left[\frac{L^2}{4\pi}\right] = \frac{1}{4\pi} E[L^2]$$

$$E[A'] = 572,96 \text{ cm}^2$$

b) si el $A \sim \text{Exp}(\lambda)$ $E[A] = 15 \text{ cm}^2$

calcular perímetro $\rightarrow$ media

$$\frac{1}{\lambda} = 15 \text{ cm}^2$$

$$A \sim \text{Exp}(1/15)$$

$$\lambda = 1/15$$

$$A = \pi r^2$$

Pongamos L en función de A

$$E[g(a)] = \int_0^\infty g(a) \cdot f_a(a) da \quad g(r) = \underbrace{\sqrt{\frac{A}{\pi}} \cdot 2\pi}_{\sqrt{A \cdot 4\pi}} = L \quad L = \sqrt{\frac{A}{\pi}} \cdot 2\pi$$

$$E[g(a)] = \int_0^\infty \sqrt{A \cdot 4\pi} \cdot \frac{1}{15} \cdot e^{-\frac{1}{15}a} da$$

$$f_r(r) = \lambda e^{-\lambda r}$$

$$= 2\sqrt{\pi} \int_0^\infty \sqrt{a} \cdot e^{-\frac{1}{15}a} \cdot \frac{1}{15} da$$

gamma

$$\frac{\lambda^v}{\Gamma(v)} x^{v-1} e^{-\lambda x}$$

$$\lambda = 1/15$$

$$x = a$$

$$\sqrt{a} = a^{v-1}$$

$$v-1 = 1/2$$

$$v = 3/2$$

$$\Gamma(3/2) = \Gamma(1 + 1/2) =$$

$$\frac{1}{2} \Gamma(1/2) = \frac{1}{2} \sqrt{\pi}$$

$$\lambda^v = \left(\frac{1}{15}\right)^{3/2}$$

$$= \frac{2\sqrt{\pi} \Gamma(3/2)}{15 \left(\frac{1}{15}\right)^{3/2}} \int_0^\infty \sqrt{a} e^{-a/15} \left(\frac{1}{15}\right)^{3/2} \frac{1}{\Gamma(3/2)} da$$

$$E[g(a)] = \frac{2\sqrt{\pi} \cdot \frac{1}{2} \pi}{\left(\frac{1}{15}\right)^{3/2} \cdot 15} = 21.6 \text{ cm}^2$$

$$\Gamma(\tau+1) = \tau \Gamma(\tau)$$

$$\Gamma(1/2) = \sqrt{\pi}$$

VARIANZA = como se dispersan los valores alrededor del promedio $E[X]$

$$VAR(X) = E[(X - \mu)^2]$$

$$VAR(X) = E[X^2] - E^2[X]$$

+var : + alejados
-var : + cerca

$$E[X^2] = \int_{-\infty}^{+\infty} x^2 f_X(x) dx$$

DISCRETO

$$Var(X) = \sum_{x \in A} (x - \mu)^2 p_X(x)$$

CONTINUO

$$Var(X) = \int_{-\infty}^{+\infty} (x - \mu)^2 f_X(x) dx$$

PROP:

- $VAR(c \cdot x) = 0$

- $VAR(X) \geq 0$

- $VAR(x + c \cdot x) = VAR(x)$

- $VAR(ax) = a^2 VAR(x)$

- $VAR(X+Y) = VAR(X) + VAR(Y) + 2COV(X,Y)$

- $VAR(X-Y) = VAR(X) + VAR(Y) - 2COV(X,Y)$

$$COV[X,Y] = E[XY] - E[X]E[Y]$$

$$V[ax+by] = a^2 V[X] + b^2 V[Y] + 2ab COV[X,Y]$$

PROP:

- X y Y son ind $\rightarrow COV(X,Y) = 0$

- $COV(aX,Y) = a COV(X,Y)$

- $COV(X,Y) = COV(Y,X)$

- $COV(X+Y,Z) = COV(X,Z) + COV(Y,Z)$

- $COV(X,X) = E[X^2] - E^2[X] = VAR(X) \geq 0$

ESPERANZA DE VECTORES

DISCRETO

$$E[g(X,Y)] = \sum_{(x,y) \in A} g(x,y) P_{XY}(x,y)$$

$$A = \{(x,y) \in \mathbb{R}^2 / P(X=x, Y=y) > 0\}$$

CONTINUO

$$E[g(X,Y)] = \iint_{\mathbb{R}^2} g(x,y) f_{XY}(x,y) dx dy$$

$$E[XY] = E[X]E[Y]$$

si son ind. $\nRightarrow$

$\downarrow$
SOB $\neq$ rectangulo
NO son ind.

ERA MIXTA

$$T^* = \begin{cases} T & \text{si } T < 1 \\ 1 & \text{si } T \geq 1 \end{cases}$$

$$F_T(t) = \begin{cases} 1 - e^{-t} & \text{si } t > 0 \\ 0 & \text{si } t \leq 0 \end{cases}$$

3.11 Sea T^* v.a. del 2.12. $\rightarrow T^* = \min(T, 1)$

$$E[T^*] \text{ y } \text{var}[T^*]$$

T : duraci3n en aros del tiempo de trabajo simfallas

$$E[T^*] = \int_0^1 t^* e^{-t^*} dt^* + 1e^{-1} = 0,632$$

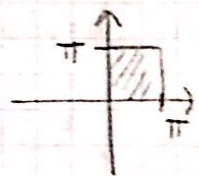
$$V(T^*) = E[T^{*2}] - E^2[T^*]$$

$$= \int_0^1 t^{*2} e^{-t^*} dt + 1e^{-1} - 0,632^2 = 0,129$$

3.12 X, Y son v.a. ind y UNIFORMES sobre $(0, \pi)$ → CONTINUA

$$E\left[\frac{x \sin(xy)}{g(xy)}\right] = \iint_{\mathbb{R}^2} g(xy) f_{XY}(x, y) dx dy$$

como son ind



$$f_X(x) f_Y(y) = f_{XY}(x, y)$$

$$\frac{1}{\pi} \cdot \frac{1}{\pi} = f_{XY}(x, y)$$

$$\frac{1}{\pi^2} = f_{XY}(x, y)$$

$$E[X \sin(XY)] = E[X] \cdot E[\sin(XY)] \quad \text{por ser ind}$$

$$E[X] = \frac{\pi}{2} \rightarrow \text{por ser unif}$$

$$E[\sin(XY)] = \int_0^\pi \int_0^\pi \sin(xy) \frac{1}{\pi^2} dx dy = \frac{1}{\pi^2} \int_0^\pi \left[-\cos(xy) \frac{1}{y} \Big|_0^\pi \right] dy$$

$$= -\frac{1}{\pi^2} \int_0^\pi \frac{\cos(\pi k)}{y} - \frac{1}{y} dy = 7,40 \times 10^{-3}$$

$$E[X \sin(XY)] = \frac{7,40 \times 10^{-3} \pi}{2} = 0,012$$

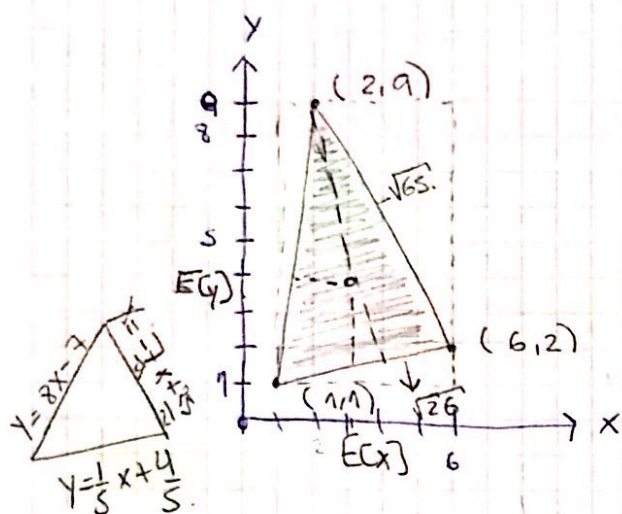
3.13

Seca (X,Y)

v.a $\sim U(\Delta)$

$\Delta: (1,1)(6,2)(2,9)$

$E[X] \quad Y \quad E[Y]$



$$\text{Area} = \frac{b \cdot h}{2} = \frac{39}{2}$$

$$h^2 = 1^2 + 5^2$$

$$h = \sqrt{26} = b$$

$$h^2 = 7^2 + 4^2$$

$$h = \sqrt{65}$$

$$h^2 = \text{ALT}^2 + \left(\frac{b}{2}\right)^2$$

$$65 = \text{ALT}^2 + \left(\frac{\sqrt{26}}{2}\right)^2$$

$$\sqrt{65 - \frac{26}{4}} = \text{ALT}$$

$$7.165 \text{ m} = \text{ALT}$$

$$E[X] = \iint_{\mathbb{R}^2 \rightarrow \text{supp}(X,Y)} x \cdot f_{X,Y} dx dy = \int_{-\infty}^{+\infty} x f_X(x) dx$$

$$E[Y] = \iint_{\mathbb{R}^2 \rightarrow \text{supp}(X,Y)} y f_{X,Y} dx dy = \int_{-\infty}^{+\infty} y f_Y(y) dy$$

$$f_{X,Y} = \frac{1}{\text{area}}$$

$$f_{X,Y} = \frac{2}{39}$$

$$E[X] = \int_1^2 \int_{\frac{1}{5}x + \frac{4}{5}}^{8x-7} x \cdot \frac{2}{39} dx dy + \int_2^6 \int_{\frac{1}{5}x + \frac{4}{5}}^{-7/4x + \frac{25}{2}} x \cdot \frac{2}{39} dx dy$$

$$E[X] = 3$$

$$E[Y] = \int_1^2 \int_{\frac{1}{5}x + \frac{4}{5}}^{8x-7} y \cdot \frac{2}{39} dx dy + \int_2^9 \int_{8x-7}^{-7/4x + \frac{25}{2}} y \cdot \frac{2}{39} dx dy =$$

$$E[Y] = \int_1^2 \frac{2}{39} \left[\frac{y^2}{2} \right]_{\frac{1}{5}x + \frac{4}{5}}^{8x-7} dx + \int_2^9 \frac{2}{39} \left[\frac{y^2}{2} \right]_{8x-7}^{-7/4x + \frac{25}{2}} dx =$$

$V(x) = 1 \rightarrow \text{var máxima}$
 $V(x) = 0 \rightarrow \text{var mínima}$

3.14) X va $\{2, 3, 4\}$ $P(X=x) = p_x$, $x \in \{2, 3, 4\}$
de media 3

a)

$$V[X] = \underbrace{E[X^2]}_{\downarrow} - \underbrace{E^2[X]}_9$$

$$\sum x^2 p_{xi} = \underbrace{x_2^2}_{2^2} \cdot p_2 + \underbrace{x_3^2}_{3^2} p_3 + \underbrace{x_4^2}_{4^2} p_4 =$$

$$V[X] = 1 = 4p_2 + 9p_3 + 16p_4 - 9$$

y se que $p_2 + p_3 + p_4 = 1$

$$p_2 = 1 - p_3 - p_4$$

$$E[X] = 2p_2 + 3p_3 + 4p_4 = 3$$

$$\therefore \boxed{p_3 = 1 - 2p_4} \rightarrow \boxed{p_2 = p_4}$$

$$1 = 4p_4 + 9(1 - 2p_4) + 16p_4 - 9$$

$$10 = 4p_4 + 9 - 18p_4 + 16p_4$$

$$1 = 2p_4 \rightarrow \boxed{1/2 = p_4} \rightarrow p_2 = 1/2 \quad p_3 = 0$$

b) $V[X] = 0 \rightarrow$

$$0 = 4p_4 + 9(1 - 2p_4) + 16p_4$$

$$0 = 4p_4 + 9 - 18p_4 + 16p_4$$

$$\boxed{0 = p_4} \quad \boxed{p_2 = 0} \quad \boxed{p_3 = 1}$$

3.15) Sea X uma V.A. $\sim U(8, 10)$

$$E[a] = a$$

a) $Y = 2(X - 1) = 2X - 2$

$$E[X] = \frac{8+10}{2}$$

$$E[Y] = E[2X - 2] = \underset{\substack{\uparrow \\ \text{PROP}}}{2} E[X] - \underset{\substack{\downarrow \\ 2}}{E[2]} =$$

$$E[Y] = 18 - 2 = 16$$

$$V[Y] = E[Y^2] - E^2[Y]$$

$$V[2X - 2] = E[(2X - 2)^2] - \underbrace{E^2[2X - 2]}_{16^2}$$

$$V[Y] = E\left[4X^2 - \frac{2(2X)(2)}{8X} + 4\right] - 256$$

$$V[Y] = 4E[X^2] - 8E[X] + E[4] - 256$$

$$V[Y] = 4 \cdot \frac{244}{3} - 8 \cdot 9 + 4 - 256$$

$$V[Y] = 1133$$

$$VAR[X] = E[X^2] - E^2[X] \\ \left(\frac{10-8}{12}\right)^2 = E[X^2] - 81$$

$$E[X^2] = 81 + \frac{41}{123}$$

$$E[X^2] = 244/3$$

b) $Y = 2X^2 + 1$

$$E[Y] = E[2X^2 + 1] = 2E[X^2] + E[1]$$

$$E[Y] = \frac{491}{3}$$

c) $Y = 2(X-1)(X-3) = (2X-2)(X-3) = 2X^2 - \frac{-8X}{6X-2X} + 6$

$$= 2E[X^2] - 8E[X] + E[6] =$$

$$= 2 \cdot \frac{244}{3} - 8 \cdot 9 + 6$$

$$E[Y] = \frac{290}{3}$$

Para hallar $E_{\max}$ o $E_{\min}$,
 veo donde se anula y evalúo
 en ese punto

d) $E[(X-c)^2] \rightarrow$ Hallar c para $E_{\min}$.

$$E[(X-c)^2] = 0$$

$$E[X^2 - 2Xc + c^2] = E[X^2] - 2cE[X] + E[c^2]$$

$$0 = \frac{244}{3} - 2c \cdot 9 + c^2$$

$$0 = c^2 - 18c + \frac{244}{3} \quad \begin{cases} c_1 = 9 \\ c_2 = 9 \end{cases}$$

$$\text{En } c=9.$$

$$E[(X-9)^2] = \frac{1}{3}$$

↓
mínima.

en $c=8$ y $c=10$ } "los alrededores"

$$E(X-8)^2 = 4/3$$

$$E(X-10)^2 = 4/3$$

e) Hallar a y b tal que $aX + b$ tenga media 0 y desvío 1

$$E[aX+b] = 0$$

$$V[aX+b] = 1$$

$$aE[X] + E[b] = 0$$

$$V[aX+b] = E[(aX+b)^2] - \underbrace{E^2[aX+b]}_0$$

$$a \cdot 9 + b = 0$$

$$V[aX+b] = E[a^2X^2 + 2aXb + b^2] - c$$

$$\boxed{b = -9a}$$

$$a^2[E[X^2]] + 2abE[X] + b^2 = 1$$

$$a^2 \cdot \frac{244}{3} + 18ab + b^2 = 1$$

$$a^2 \cdot \frac{244}{3} - 162a^2 + 81a^2 = 1$$

$$\begin{cases} a = \sqrt{3} & b = -9\sqrt{3} \\ a = -\sqrt{3} & b = 9\sqrt{3} \end{cases}$$

$$\boxed{a = \pm \sqrt{3}}$$

$$\boxed{b = \pm 9\sqrt{3}}$$

3.16

$$W = rV^2$$

$$r = \text{cte.}$$

Voltage $\sim$ normal

$$E[W] \text{ con } r=3$$

media 6 y var 1.

$$E[3V^2] = 3 E[V^2]$$

$$\mu = 6$$

$$\sigma^2 = 1$$

$$\text{Var}[V] = E[V^2] - E^2[V]$$

$$1 + 36 = E[V^2]$$

$$37 = E[V^2]$$

$$\rightarrow E[3V^2] = 3 \cdot 37 = \boxed{111}$$

3.17

X_1 y $X_2 \sim \text{Bernoulli}$

$\text{Ber}(p_1)$ y $\text{Ber}(p_2)$

a) $\text{cov}(X_1, X_2) = 0 \rightarrow$ prob total.

$$\text{cov}(X_1, X_2) = E[X_1 X_2] - E[X_1] E[X_2] \text{ also ind.}$$

$$= E[X_1] E[X_2] - E[X_1] E[X_2] = 0 \quad \checkmark \quad E(XY) = E(X)E(Y)$$

b) si $p_1 = p_2 = 0,7$ $\text{cov}(X_1, X_2) = 0,1$ hallar $P(Y_1, Y_2)$
 siendo $Y_1 = X_1(1-X_2)$ $Y_2 = X_2(1-X_1)$

all a) $\text{COV}(Y_1, Y_2) = E[X_1 X_2] - E[X_2] E[X_1]$

$$\text{COV}(X_1, X_2) = \sum_{X_1=0}^1 \sum_{X_2=0}^1 X_1 X_2 P_{X_1 X_2}(X_1, X_2) - p_1 p_2$$

si $\text{cov}(X_1, X_2) = 0 \rightarrow P_{X_1 X_2} = p_1 p_2$

según b

$$\rho(Y_1, Y_2) = \frac{\text{COV}(Y_1, Y_2)}{\sqrt{V(Y_1)} \sqrt{V(Y_2)}} =$$

$$\rho(Y_1, Y_2) = \frac{E[Y_1 Y_2] - E[Y_1] E[Y_2]}{\sqrt{V(Y_1)} \sqrt{V(Y_2)}}$$

$$\rho(Y_1, Y_2) = \frac{\text{COV}(Y_1, Y_2)}{\sqrt{V(Y_1)} \sqrt{V(Y_2)}}$$

↓
coeficiente de correlación

Y_2/Y_1	0	1	P_{Y_2}
0	0,78	0,11	0,89
1	0,11	0	0,11
P_{Y_1}	0,89	0,11	

$$\text{COV}(Y_1, Y_2) = P_{Y_1 Y_2} - P_{Y_1} P_{Y_2}$$

↑ $P_{X_1 X_2} = P_{Y_1 Y_2} + \text{COV}$

$$P_{Y_1 Y_2}(0,0) = P_{X_1 X_2}(0,0) + P_{X_1 X_2}(1,1)$$

$$Y_1=0 \rightarrow X_1 \cdot (1-X_2)$$

$$Y_2=0 \rightarrow \oplus$$

$\begin{cases} X_1=0 \\ X_2=0 \end{cases}$
 $\begin{cases} X_1=1 \\ X_2=1 \end{cases}$

 $\oplus$

$$P_{Y_1 Y_2}(0,0) = \overbrace{(1-0,7)}^{P_{X_1=0}} \overbrace{(1-0,7)}^{P_{X_2=0}} + 0,11 + 0,11 + \underbrace{0,7 \cdot 0,7}_{P_{X_1=1} P_{X_2=1}} = 0,78$$

↑ intersección

$$P_{Y_1 Y_2}(1,0) = P_{X_1 X_2}(1,0) = P_{X_1=1} \cdot P_{X_2=0} = 0,11$$

$$\text{COV}(Y_1, Y_2) = P_{X_1 X_2}(X_1, X_2) - P_{Y_1} P_{Y_2} = 0$$

$$P_{X_1 X_2} = P_{Y_1} P_{Y_2}$$

$$\text{COV}(Y_1, Y_2) = \overbrace{P_{Y_1 Y_2}}^{E[Y_1 Y_2]} - \overbrace{P_{Y_1} P_{Y_2}}^{E[Y_1] E[Y_2]} = -0,0121$$

$$Y_1 \sim B(0,11)$$

$$Y_2 \sim B(0,11)$$

$$\rho(Y_1, Y_2) = \frac{-0,0121}{\sqrt{0,11} \sqrt{0,89}} = \checkmark$$

⇒
(enclosed)

NOTA

otra forma .

$$\sqrt{V(Y_1)} = \sigma_Y$$

$$\rho(Y_1, Y_2) = \frac{\text{cov}(Y_1, Y_2)}{\sigma_{Y_1} \sigma_{Y_2}} = \frac{E[Y_1 Y_2] - E[Y_1]E[Y_2]}{\sqrt{E[Y_1^2] - E^2(Y_1)} \sqrt{E[Y_2^2] - E^2(Y_2)}}$$

$$E[Y_1] = E[X_1] - E[X_1 X_2] = 0,7 - 0,49 = 0,21$$

$$E[Y_2] = E[X_2] - E[X_1 X_2] = 0,7 - 0,49 = 0,21$$

$$\text{cov}(X_1, X_2) = 0,1$$

$$V[X_1] = 0,21 = V[X_2]$$

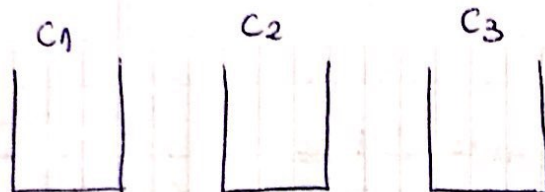
↓

$$P(1-P)$$

no//

$$E[Y_1 Y_2] = P_{Y_1} P_{Y_2} = 0$$

3.18.



X_i : cantidad de bolas en C_i $X_P = 0, 1, 2, 3$

N : cantidad de urnas con alguna bola. $N = 1, 2, 3$

a)

N / X_1	0	1	2	3	$P_N(m)$
1	$2/27$	0	0	$1/27$	$3/27$
2	$6/27$	$6/27$	$6/27$	0	$18/27$
3	0	$6/27$	0	0	$6/27$
$P_X(x)$	$8/27$	$12/27$	$6/27$	$1/27$	1

$$\# \Omega = 3^3 = 27$$

→ todas en una urna

→ todas en 2 urnas

→ todas en 3 urnas

$$P(X_P, N) =$$

$$P(0, 1) = 2/27 \quad P(1, 1) = 0 \quad P(2, 1) =$$

↓
0 bolas en X_1
3 en → todas X_2
todas X_3

2 en X_1
1 todo

$$P(0, 2) = 6/27$$

0 en X_1
2 urnas →

X_1
0
0
0
 X_2
 $b_1 b_2$
 $b_1 b_3$
 $b_2 b_3$
 X_3
 b_3
 b_2
 b_1

$$P(2, 2) =$$

X_1 X_2 X_3
 $b_1 b_2$ b_3 b_3 } X_3
 $b_1 b_2$ b_3 b_3

$$P(1, 2) =$$

1 bola X_1
2 condgo

X_1 X_2 X_3
 b_1 $b_2 b_3$ 0 } X_2
 b_2 $b_1 b_3$ 0
 b_3 $b_2 b_1$ 0

$$P(3, 2) = 0$$

$$P(1, 3) = 6/27$$

X_1 X_2 X_3
 b_1 b_2 b_3 } X_3
 b_1 b_2 b_3

$$E[N] = \sum N \cdot P_N = \overset{\text{prob total}}{\uparrow} \overset{N}{1} \cdot \frac{3}{27} + \overset{N}{2} \cdot \frac{18}{27} + \overset{N}{3} \cdot \frac{6}{27} = \boxed{\frac{19}{9}}$$

$X=1 \rightarrow$ todos en 1.

$$\text{VAR}[N] = E[N^2] - E^2[N] = \boxed{\frac{26}{81}}$$

$$E[N^2] = 1^2 \cdot \frac{3}{27} + 4 \cdot \frac{18}{27} + 9 \cdot \frac{6}{27} = \frac{43}{9}$$

NOTA

$$\text{COV}(N, X_1) = E[N, X_1] - E[N] E[X_1] =$$

$$\boxed{\text{COV}(N, X_1) = 0.}$$

$$E[N, X_1] = \sum N \cdot X_1 \cdot P_{N, X} = 2 \cdot 1 \cdot \frac{6}{27} + 3 \cdot 1 \cdot \frac{6}{27} + 2 \cdot 2 \cdot \frac{6}{27} +$$

$$3 \cdot 1 \cdot \frac{1}{27} = \frac{19}{9}.$$

$$E[X_1] = \sum X \cdot P_X = 1 \cdot \frac{12}{27} + 2 \cdot \frac{6}{27} + \frac{1}{27} \cdot 3 + 0 \cdot \frac{8}{27} = 1.$$

$$E[N, X_1] = \sum N \cdot X_1 \cdot \underbrace{P_{N, X_1}}_{\text{conjunta}}.$$

$$E[X_1] = \sum X_1 \cdot \underbrace{P_{X_1}}_{\text{marginal}}$$

$$E[N] = \sum N \cdot \underbrace{P_N}_{\text{marginal}}$$

b)

$$\text{Si fueran ind. } P(X_i, N_i) = P(X_i) P(N_i)$$

$$P(X_1 = 0, N = 1) = P(X_1 = 0) P(N = 1)$$

$$\frac{2}{27} \neq 0 \cdot \frac{1}{27}$$

$\downarrow$
 $\frac{3}{27}$ $\downarrow$ $\frac{3}{27} \rightarrow$ comp. prob.

Puede ocurrir.

$$\text{COV}(X_1, N) = 0$$

y no son ind.

c) $\text{COV}(X_i, X_j)$

$$1 \leq i \leq j \leq 3$$

$$\text{COV}(X_1, X_1) = \text{VAR}(X_1)$$

$$= E[X_1^2] - E^2[X_1].$$

$$1 \leq i \leq 3$$

$$1 \leq j \leq 3$$

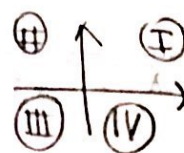
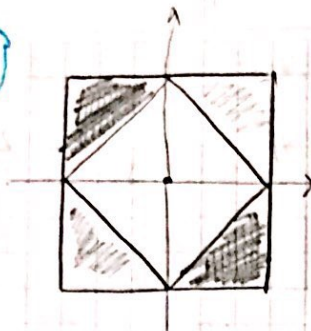
11	21	31
12	22	32
13	23	33

$\Rightarrow$ muy largo.

$$E[X_1 X_2] = \sum X_1 X_2 \cdot P_{X_1 X_2} \dots$$

3.19

a)



- $f_{xy} = 1$
- ◐ $f_{xy} = 1/2$
- $f_{xy} = 3/2$

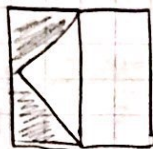
La cov > 0 mientras $\oplus$ puntos halla en $\oplus$ y $\ominus$

$$\text{COV}(X, Y) = E[XY] - E[X]E[Y]$$

$$\text{COV}(X, Y) < 0$$

pg la densidad de mayor valor está en los cuadrantes $\oplus$ y $\ominus$ q's son neg

b)



$\text{COV}(X, Y) = 0$ pg se compensan al estar en cuadr $\oplus$ y $\ominus$

c)

cuando $\oplus > 0$ y $\ominus < 0$

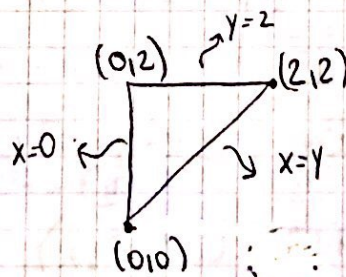
$$\text{pero } |\text{COV}_2(X, Y)| < |\text{COV}_1(X, Y)|$$

↓ tiene menos área

3.20

X, Y vector aleat.
 $\sim U(\Delta)$

Continua



$$\text{Área } \Delta = 2$$

$$f_{xy} = \frac{1}{\text{Área}}$$

a) $\text{COV}(X, Y) = E[XY] - E[X]E[Y] = 1/9$

$$E[XY] = \iint_{\mathbb{R}^2} xy f_{xy} = \int_0^2 \int_0^2 \frac{xy}{2} dx dy = 1$$

$$E[X] = \int_0^2 \int_0^2 x \cdot \frac{1}{2} dx dy = \frac{2}{3} \quad E[Y] = \int_0^2 \int_0^2 y \cdot \frac{1}{2} dx dy = \frac{4}{3}$$

NOTA

$$b) \text{VAR}[x+y] = E[(x+y)^2] - E^2[x+y] =$$

$$= x^2 + 2xy + y^2.$$

$$\text{VAR}[x+y] = E[x^2] + 2E[xy] + E[y^2] - E^2[x+y] =$$

↓
móltorugo

$$E[x^2] = \int_0^2 \int_0^2 \frac{x^2}{2} dx dy = \frac{2}{3}$$

$$E[y^2] = \int_0^2 \int_0^2 \frac{y^2}{2} dx dy = 2$$

$$\text{VAR}[x+y] = \text{VAR}[x] + \text{VAR}[y] + 2\text{COV}[x,y]$$

$$\text{VAR}[x] = E[x^2] - E^2[x] = \frac{2}{3} - \frac{4}{9} = \frac{2}{9}$$

$$\text{VAR}[y] = 2 - \frac{16}{9} = \frac{2}{9}$$

$$\text{VAR}[x+y] = \frac{2}{9} + \frac{2}{9} + 2 \cdot \frac{1}{9} = \frac{2}{3}$$

$$c) \text{COV}(3x - y + 2, x + y) = \frac{2}{3}$$

$$\text{COV}(3x - y + 2, x) + \text{COV}(3x - y + 2, y)$$

$$\boxed{\text{COV}(2, x) = 0}$$

$$\textcircled{\oplus} \text{COV}(3x, x) - \text{COV}(y, x) + \text{COV}(2, x)$$

$$3\text{COV}(x, x) - \text{COV}(x, y) = 3V(x) - \text{COV}(x, y) = 3 \cdot \frac{2}{9} - \frac{1}{9} = \frac{5}{9}$$

$$\textcircled{\oplus} \text{COV}(3x, y) - \text{COV}(y, y) + \text{COV}(2, y) =$$

$$3\text{COV}(x, y) - V(y) = 3 \cdot \frac{1}{9} - \frac{2}{9} = \frac{1}{9}$$

Recta de Regresión

(X, Y) vector aleatorio. "conocido" la v.a. X ; se obtiene info sobre Y . "estimación"

$\hat{y} = ax + b$ tal que $E[(Y - \hat{y})^2]$ sea mínima
Recta Regresión

ESTIMACIÓN
de Y en f.
de X

$$\hat{y}(x) = E(Y) + \frac{\text{COV}(X, Y)}{V(X)} (X - E(X))$$

↓
Es la mejor estimación
lineal de Y dada X .

3.21 Juan y Maria

X : tiempo de reacción de Juan
 Y : tiempo de reacción de Maria

$X, Y \sim U(0, 2)$
son ind.

→ W : tiempo de reacción del ganador.
→ L : tiempo de reacción del perdedor.

a) Recta de $[L|W]$ "L dado W"

$$L = aW + b$$

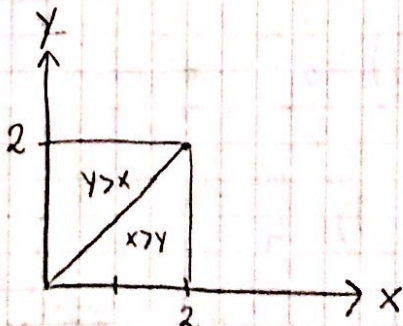
$$W = \min\{X, Y\}$$

$$\begin{cases} X & \text{si } X < Y \\ Y & \text{si } X \geq Y \end{cases}$$

$$L = \max\{X, Y\} \begin{cases} X & \text{si } X > Y \\ Y & \text{si } X \leq Y \end{cases}$$

$$\hat{L}(W) = E[L] + \frac{\text{COV}(W, L)}{V(W)} (W - E[W])$$

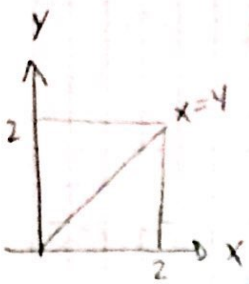
$$\hat{L}(W) = E[h(X, Y)] + \frac{\text{COV}[g, h]}{V(g(X, Y))} (g(X, Y) - E[g]) \quad \left\{ \begin{array}{l} \text{llamamos} \\ W = g(X, Y) \\ L = h(X, Y) \end{array} \right.$$



$$f_{XY}(X, Y) = \frac{1}{\text{área}} = \frac{1}{4} \mathbb{1}_{\{(X, Y) \in [0, 2]^2\}}$$

$$E[g(x,y)] = \iint_{\mathbb{R}^2} g(x,y) \cdot f_{x,y} \, dx \, dy$$

↳ como temos uma função suportada dividida em dois integrais



$$\int_0^2 \int_x^2 x \cdot \frac{1}{4} \, dy \, dx \, || \{x < y\} + \int_0^2 \int_0^x y \cdot \frac{1}{4} \, dy \, dx \, || \{x \geq y\}$$

$g(x,y)$

↳ x $x < y$
↳ y $x \geq y$

$$\int_0^2 \left. \frac{x}{4} \cdot y \right|_x^2 \, dx + \frac{1}{4} \int_0^2 \left. \frac{y^2}{2} \right|_0^x \, dx$$

$$\int_0^2 \frac{x}{4} (2-x) \, dx + \frac{1}{4} \int_0^2 \frac{x^2}{2} \, dx =$$

$$\frac{1}{2} \cdot \frac{x^2}{2} - \frac{x^3}{3} \cdot \frac{1}{4} \Big|_0^2 + \frac{1}{4} \left(\frac{x^3}{3 \cdot 2} \right) \Big|_0^2 =$$

$$\frac{4}{4} - \frac{8}{12} + \frac{1}{4} \left(\frac{8}{6} \right) = \frac{2}{3} = E[g(x,y)] \rightarrow E[W]$$

$$E[h(x,y)] = \int_0^2 \int_x^2 x \cdot \frac{1}{4} \, dy \, dx \, || \{x \geq y\} + \int_0^2 \int_0^x y \cdot \frac{1}{4} \, dy \, dx \, || \{x < y\}$$

$h(x,y)$

↳ x $x \geq y$
↳ y $x < y$

$$E[h(x,y)] = \frac{1}{4} \int_0^2 x^2 \, dx + \frac{1}{4} \int_0^2 \left. \frac{y^2}{2} \right|_x^2 \, dx \rightarrow \left(\frac{2}{4} \right) - \frac{x^2}{2}$$

$$E[h(x,y)] = \frac{1}{4} \left. \frac{x^3}{3} \right|_0^2 + \frac{1}{4} \left[2x - \frac{x^3}{6} \right] \Big|_0^2$$

$$= \frac{1}{4} \left(\frac{8}{3} \right) + \frac{1}{4} \left(4 - \frac{8}{6} \right) = \frac{2}{3} + \frac{2}{3} = \frac{4}{3} = E[L]$$

$$\text{Var}(W) = E[W^2] - E^2[W] = \frac{1}{3} - \left(\frac{2}{3} \right)^2 = \frac{2}{9}$$

$$E[W^2] = \int_0^2 \int_x^2 \frac{x^2}{4} \, dy \, dx + \int_0^2 \int_0^x \frac{y^2}{4} \, dy \, dx =$$

$$E[W^2] = \int_0^2 \frac{x^2}{4} (2-x) \, dx + \frac{1}{4} \int_0^2 \frac{x^3}{3} \, dx = \frac{1}{4} \left(\frac{2x^3}{3} - \frac{x^4}{4} \right) \Big|_0^2 + \frac{1}{12} \cdot \frac{x^4}{4} \Big|_0^2$$

NOTA

$$\frac{1}{4} \left(\frac{16}{3} - \frac{16}{4} \right) + \frac{1}{4} \left(\frac{16}{12} \right) = \frac{1}{3} + \frac{1}{3} = \frac{2}{3}$$

$$\text{cov}(g, h) = E[gh] - E[h]E[g] = 1 - \left(\frac{4}{3} \cdot \frac{2}{3} \right) = \frac{1}{9}$$

$$E[gh] = E[xy] = \int_0^2 \int_0^2 \frac{xy}{4} dx dy = \frac{1}{4} \int_0^2 \frac{x^2}{2} \Big|_0^2 y dy$$

$$= \frac{1}{4} \left(\frac{4}{2} \right) \frac{y^2}{2} \Big|_0^2 = \frac{1}{4} \cdot 2 \cdot 2 = 1$$

$\text{if } x < y$
 $gh = xy$
 $\text{if } x \geq y$
 $gh = xy$

$\left. \begin{array}{l} \text{if } x < y \\ \text{if } x \geq y \end{array} \right\} \begin{array}{l} \text{value of } xy \\ \text{is same} \end{array}$

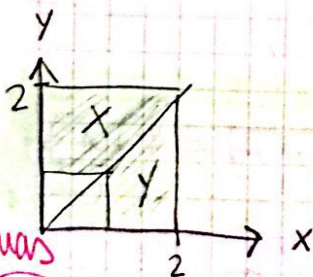
$$\hat{L}(w) = \frac{4}{3} + \frac{\frac{1}{9}}{\frac{2}{9}} \left(w - \frac{2}{3} \right) = \frac{4}{3} + \frac{1}{2} \left(w - \frac{2}{3} \right)$$

$$\frac{1}{2} \left(-\frac{2}{3} \right) = -\frac{1}{3}$$

$$\hat{L}(w) = \frac{1}{2} w + 1$$

b)

$$E[W | L > 1] = \frac{E[W \cdot \mathbb{1}_{\{L > 1\}}]}{P(L > 1)} = \frac{\int_1^2 \int_0^y \frac{x}{4} + \int_1^2 \int_0^x \frac{y}{4} dy dx}{1 - \underbrace{P(L \leq 1)}_{F_{xy}(1)}}$$



eu continuous

$$P(L > 1) = 1 - P(L \leq 1)$$

$$P(L \leq 1) = \iint_{\text{Area } L \leq 1} f_{xy} dx dy$$

Area total = 1

$$\iint_A \frac{1}{\text{Area}} dx dy = \frac{1}{4} \text{ Area } (L \leq 1) = \frac{1}{4}$$

$$1 - \frac{1}{4} = \frac{3}{4}$$

$$\frac{7/24 + 7/24}{3/4}$$

$$\boxed{E[W | L > 1] = 7/9}$$

3.22

$$f_{xy}(x,y) = \frac{5}{8\pi} e^{-\frac{25}{32}(x^2 - \frac{6}{5}xy + y^2)}$$

$$= \frac{1}{\pi \frac{8}{5}} e^{-\frac{25}{32}((y - \frac{3}{5}x)^2 + \frac{16}{25}x^2)}$$

↳ factorizo para buscar 2 normales

$$= \frac{1}{\sqrt{\pi}} \frac{5}{8} e^{-\frac{(y - \frac{3}{5}x)^2}{32/25}} \cdot \frac{1}{\sqrt{\pi}} e^{-\frac{1}{2}x^2} =$$

$$= \underbrace{\frac{1}{\sqrt{2\pi}} \frac{5}{8} e^{-\frac{1}{2} \frac{(y - \frac{3}{5}x)^2}{16/25}}}_{f_{y|x=x}} \cdot \underbrace{\frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}x^2}}_{f_x}$$

$f_{y|x=x}$

$$X \sim N(\frac{3}{5}x, \frac{16}{25})$$

↳ no estandar

f_x

↓

$$X \sim N(0,1) \rightarrow \text{estandar}$$

$$E[Y|X] = \boxed{\frac{3}{5}x = \varphi(x)}$$

como mi función φ (regresión) es lineal, esa ya es la ~~función~~ Recta de Regresión

3.23

$$X \text{ va } \geq 0, E[X] = 15$$

$$P(X \geq 60) \leq 0.25$$

Por MARKOV

$$P(X > a) \leq \frac{E[X]}{a}$$

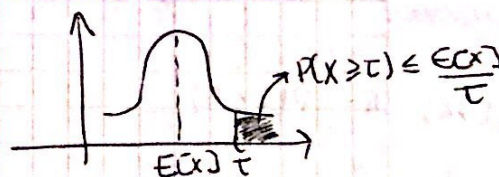
$$P(X \geq 60) \leq \frac{15}{60}$$

$$P(X \geq 60) \leq 0.25 \checkmark$$

Desigualdad de
MARKOV

Sea $X \geq 0$, con $E[X]$ finita
 $\forall \tau > 0$,

$$P(X \geq \tau) \leq \frac{E[X]}{\tau}$$



3.24 X v.a $E[X] = 10$ $V(X) = 15$

$$P(5 < X < 15) \geq 0.4$$

Por Chebyshev

$$P(|X - \mu| > \varepsilon) \leq \frac{V(X)}{\varepsilon^2}$$

$$P(|X - 10| \leq 5) \geq 1 - \frac{V(X)}{\varepsilon^2} = 1 - \frac{15}{5^2}$$

$$P(5 \leq X \leq 15) \geq 0.4$$

$$|X - 10| \begin{cases} X - 10 \\ -X + 10 \end{cases}$$

$$X - 10 \leq 5 \Rightarrow X \leq 15$$

$$-X + 10 \leq 5 \Rightarrow 5 \leq X$$

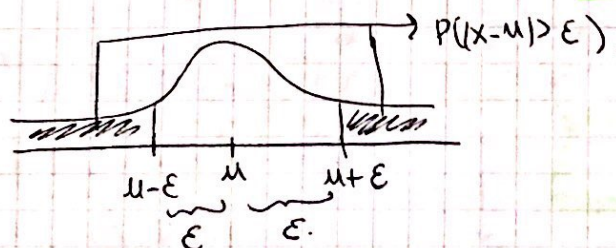
Desigualdad de Chebyshev

Sea X v.a cualquiera, con $E[X]$ y $V(X)$ finitos.
 $E[X] = \mu$ $V(X) = \sigma^2$ $\forall \varepsilon > 0$

$$P(|X - \mu| > \varepsilon) \leq \frac{V(X)}{\varepsilon^2}$$

$$P(|X - \mu| \leq \varepsilon) \geq 1 - \frac{V(X)}{\varepsilon^2}$$

Equivalentes



3.25 X : Peso paquete de café $X \sim \text{normal}(500g, 6)$

¿Cómo deben ser los valores de σ , para tener una seg del 99 % de que el peso promedio de 100 paquetes no se desvíe en más de 10g de 500g?

$$6 / P(490 < \bar{x} < 510) = 0,99$$

como varia
X: peso

99% → seguridad

NORMAL ~ (500, 6)

$$E[\bar{x}] = \frac{1}{100} \sum_{i=1}^{100} E[x_i] = 500$$

siendo

$$\bar{x} = \frac{\sum_{i=1}^{100} x_i}{100}$$

Peso promedio
de 100 paquetes

$$\sigma = V[\bar{x}] = \frac{1}{100^2} \sum_{i=1}^{100} V(x_i) = \frac{6^2}{100}$$

$$Z = \frac{X - \mu}{\sigma} \rightarrow \text{para estandarizar}$$

$$Z = \frac{X - 500}{6^2/100} \rightarrow X = Z \cdot \frac{6^2}{100} + 500$$

$$P\left(\frac{490 - 500}{\frac{6^2}{100}} < Z < \frac{510 - 500}{\frac{6^2}{100}}\right) = 0,99$$

$$P\left(-\frac{1000}{6^2} < Z < \frac{1000}{6^2}\right) = 2\Phi\left(\frac{1000}{6^2}\right) - 1 \approx 0,99$$

$$\Phi\left(\frac{100}{6^2}\right) = 0,995 \quad \text{guía 8.}$$

$$\frac{100}{6^2} = \Phi^{-1}(0,995)$$

$$6 \approx 6,23$$

SP NO

chequear

$$X - 500 \leq 10$$

$$X \leq 510$$

$$-X + 500 \leq 10$$

$$490 \leq X$$

$$P(|X - 500| \leq 10) \geq 0,99$$

$$P(|X - 500| \geq 10) \leq 1 - 0,99$$

$$\frac{V(X)}{\epsilon^2} = \frac{6^2}{\epsilon^2} \leq 0,01$$

$$\left. \begin{aligned} 1 - \frac{V(X)}{\epsilon^2} &= a \\ 1 - a &= \left(\frac{V(X)}{\epsilon^2}\right) \end{aligned} \right\} a = 0,99$$

$$6^2 \leq 10$$

$$\frac{6^2}{10^2} \leq 0,01$$

3.26 Fumadores en una población

$$0 < p < 1$$

$$P(X=1) = p$$

$$P(X=0) = 1-p$$

$$X_i = \begin{cases} 1 & \text{si } i \text{ fuma} \\ 0 & \text{si } i \text{ no fuma} \end{cases}$$

$$E[X] = p$$

$$V[X] = p(1-p)$$

→ $\begin{matrix} 1 & 0 & 0 \\ 1 & 0 & 0 \end{matrix}$
Bernoulli.

$X_i \sim \text{Be}(p)$ → prob fumador.

¿valor de m , para que la proporción de fumadores no difiera de 0,01, con prob mayor o igual a 0,95?

$$y = \frac{\sum x_i}{m} \rightarrow \text{proporción de los que fuman en muestra}$$

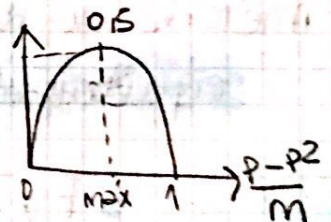
Busco $p \rightarrow V(y) = \frac{p-p^2}{m} \rightarrow \text{parabola}$

Busco sumaximo $\boxed{p=0,5}$

$$V(y) \leq \frac{0,5 - 0,5^2}{m} \quad \downarrow \text{valor max de } p$$

$\underbrace{\hspace{1cm}}_{\text{max } V(y)}$

$y = \text{proporción de fumadores}$
 $p = \text{prob de fumar}$



$$E[y] = \frac{1}{m} E[\sum x_i] = \frac{1}{m} \cdot m \cdot p = p$$

$$V[y] = \frac{1}{m^2} \sum V[x_i] = \frac{1}{m^2} m (p(1-p)) = \frac{p(1-p)}{m}$$

chequeo B.

$$P(|y-p| \leq 0,01) \geq 0,95 \quad \rightarrow \quad 1 - \frac{V(y)}{\epsilon^2}$$

$$1 - \frac{0,5 - 0,5^2}{\epsilon^2 \cdot m} \geq 0,95$$

$$\frac{1}{20} \geq \frac{1/4}{9012 \cdot m}$$

$$\boxed{m \geq 50.000}$$

IMP

$$Q = \frac{p(1-p)}{m} = \frac{p-p^2}{m}$$

$$\frac{dQ}{dp} = 0 \rightarrow \text{si suma parab.}$$

$$\downarrow \text{max}$$

$$\frac{1-0}{2} = 0,5$$

Proporcionales

aprox + V.A

$$V[X] = p(1-p)$$

$$E[X] = p$$

$y = \text{proporción de x sobre total}$

$$y = \frac{\sum x_i}{m} \rightarrow \text{total}$$

$$V[y] = \frac{1}{m^2} \sum_{i=1}^m V[x_i] = \frac{1}{m^2} \cdot m \cdot V[X]$$

$$E[y] = \frac{1}{m} \sum_{i=1}^m E[x_i] = \frac{1}{m} (m \cdot p)$$